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TRANSFORMER REGULATION

AND

ON LOAD TAP CHANGER SELECTION

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

The purpose of this standard is to highlight the requirements for Transformer voltage regulation calculations and the importance of On Load Tap Changer (OLTC) range and number of taps for the Power Transformers intended to be used in the Transmission system of Saudi Electricity Company, Kingdom of Saudi Arabia.

2.0 TRANFORMER VOLTAGE REGULATION

2.1 Introduction

Transmission power systems operate and implement standard voltages according to Industry Standards. The equipments working on these systems are given input voltages at these values within certain agreed tolerance limits. In many applications, this voltage itself may not be good enough for obtaining the best operating conditions of the loads. A transformer is interposed between the load and supply terminals in such cases. There are additional drops in the transformer due to load currents. While input voltage is the responsibilty of the supply provider, the voltage at the load is the one which the user has to worry about. It is therefore necessary to quantify the drop that takes place inside a transformer when certain load current at any power factor is drawn from the output leads. This drop is termed as voltage regulation as a ratio of the no load and load terminal voltages.

2.2 Definition of Voltage Regulation

Voltage regulation can be defined in two ways. Regulation down and regulation up. These definitions differ only in the reference voltage as illustrated below.

2.2.1 Regulation down :

This is defined as the change in terminal voltage when a load current at any power factor is applied and expressed as a fraction of the no load terminal voltage.

Expressed in symbolic form, we have,

$$\text{Regulation} = \frac{V_{nl} - V_l}{V_{nl}}$$

Where;

V_{nl} is the no load terminal voltage

V_l is the load voltage

2.2.2 Regulation up

Here the regulation is expressed as the ratio of the change in the terminal voltage, when a load at a given power factor is thrown off, and the on load voltage.

This definition can be expressed as

$$\text{Regulation} = \frac{V_{nl} - V_l}{V_l}$$

V_{nl} is the no load terminal voltage

V_l is the load voltage.

2.2.3 Normally full load regulation is of interest as the part load regulation is going to be of lower importance.

In case of transformers, both the definitions result in more or less the same value since the transformer impedance is very low and power factor of operation is quite high.
Regulation up has some advantage when it comes to its application and it depends on the load power factor.

2.3 Determination of Voltage Regulation for rated and tapping quantities

2.3.1 Voltage Regulation Formula Per IEEE C57.12.90

Percent Regulation (Lagging load)

$$= \left[\sqrt{(R + F_p)^2 + (X + q)^2} - 1 \right] \times 100\,\% \qquad \text{Eq (38.1)}$$

Where;
Fp = load Power Factor
$q = +\sqrt{1 - F_p^2}$
R = Resistance factor of Transformer $= \dfrac{\text{Load loss in kilowatts}}{\text{Rated kilovolt amperes}}$
X = Reactance factor of the Transformer $= +\sqrt{Z^2 - R^2}$
Z = Impedance of the Transformer

2.3.2 Voltage Regulation Formula per IEC 60076-8

Per Clause 7.5 of IEC 60076-8, Voltage drop shall be calculated from the formula

$$\text{Percent Regulation} = \left[\frac{U_{20}}{U_2} - 1\right] \times 100\% \qquad \text{Eq (38.2)}$$

Where U_{20} = Secondary voltage at no load
U_2 = Secondary terminal voltage with load connected.

Per equation (63) of IEC 60076-8

$$\frac{U_{20}}{U_2} = \frac{Z_L + Z_T}{Z_L} \qquad \text{Eq (38.3)}$$

Where Z_L = Load impedance in ohms
Z_T = Transformer Impedance

$$\frac{Z_L + Z_T}{Z_L} = \left[1 + \frac{S}{S_r}\left\{\frac{x_t Sin\varphi}{100} + \frac{r_t Cos\varphi}{100}\right\} + j\frac{S}{S_r}\left\{\frac{x_t Cos\varphi}{100} - \frac{r_t Sin\varphi}{100}\right\}\right] \qquad \text{(Eq 38.4)}$$

Where $\frac{S}{S_r}$ = Actual loading percentage

$Cos\ \Phi$ = Power factor

x_t & r_t are reactive and resistance components of impedance

$$\text{Hence, Percent Regulation} = \left[\frac{Z_L + Z_T}{Z_L}\right] - 1 \times 100\% \qquad \text{Eq (38.5)}$$

2.4 Sample Calculation per IEEE and IEC Guidelines

In the case of a network system where the voltage variations are dominant due to presence of reactive loads, the Transformer Voltage Regulation Calculation (voltage drop calculation) shall be related to the variation in power factor in the system. The calculation shall be as furnished below per IEEE C57.12.90 and IEC 60078-8

Example of Power Transformer Data:

a. Voltage ratio : 132/13.8 kV

b. Rating : 40/50/60/ MVA

c. Vector group : YNyno +d1

d. Rated Frequency : 60 Hz

e. Percent impedance on 60 MVA base : 25 % @ principal tap and rated frequency
26,7% @ highest tap and rated frequency
22.5% @ lowest tap and rated frequency

f. OLTC Range : -18.59% to + 10.01 % in steps of 1.43%, range + 7 to -13 taps, 21 positions (20 steps)

g. Total losses : 304 kW (Tolerance +15 % per IEC 60076) = 350 kW

2.4.1 Calculation Per IEEE C 57.12.90

$$\text{Percent Regulation} = \left[\sqrt{(R+F_p)^2+(X+q)^2}-1\right]\times 100$$

Where;

Fp = Load Power Factor

$$q = +\sqrt{1-F_p^2}$$

$$= \sqrt{1-0.8^2}$$

$$= 0.6 \text{ at } 0.8 \text{ pf}$$

R = Resistance factor of Transformer $= \dfrac{\text{Load loss in kilowatts}}{\text{Rated kilovolt amperes}}$

$$= \frac{350}{60\times 1000}$$

$$= 0.005833 \text{ ohms}$$

Value of Z in ohms(At principal tap) $= \dfrac{0.25 \text{ x } (132)^2}{60}$

$$= 72.6 \text{ ohms}$$

X = Reactance factor of the Transformer $= +\sqrt{Z^2-R^2}$

$$= \sqrt{72.6^2-0.005833^2}$$

As the value of R is very much less as compared to X, value of R can be neglected.

Hence, $X \approx Z$

Transformer regulation

at full load and 0.8 pf $= \left[\sqrt{0.8^2 + (0.25+0.6)^2} - 1\right] \times 100\%$

$= [1.167 - 1] \times 100\%$

$= 16.7\ \%$

2.4.2 Calculation Per IEC 60076-8

Percentage Voltage drop $= \left[\dfrac{U_{20}}{U_2} - 1\right] \times 100\%$

Where U_{20} = Secondary voltage at no load
U_2 = Secondary terminal voltage with load connected.

Per equation (63) of IEC 60076-8

$$\frac{U_{20}}{U_2} = \frac{Z_L + Z_T}{Z_L}$$

Where Z_L = Load impedance in ohms
Z_T = Transformer Impedance

$$\frac{Z_L + Z_T}{Z_L} = \left[1 + \frac{S}{S_r}\left\{\frac{x_t Sin\varphi}{100} + \frac{r_t Cos\varphi}{100}\right\} + j\frac{S}{S_r}\left\{\frac{x_t Cos\varphi}{100} - \frac{r_t Sin\varphi}{100}\right\}\right]$$

Where $\dfrac{S}{S_r}$ = Actual loading percentage

r_t = 0.5 % (based on 304 kW load loss)

Neglecting resistance compared to percentage impedance, $x_t = Z_t = 0.25\ \%$

Hence, regulation at 0.8 pf and full load ($Sin\varphi = 0.6$) , neglecting resistance compared to percentage impedance, shall be calculated as

$= \left[\{1 + (0.25 \times 0.6 + 0.005 \times 0.8) + j(0.25 \times 0.8 - 0.005 \times 0.6)\}\right] - 1 \times 100\%$

$= [1 + 0.154 + j0.197] - 1 \times 100\%$

$= [1.154 + j0.197] - 1 \times 100\%$

$= [1.17\text{-}1] \times 100\%$

$= 17\ \%$

As per IEC and IEEE calculations the error in % regulation is 0.3% which is neglegible.Therefore both the formula are valid.

Carrying out the calculation for 50 %, 66.67%, 75 % and 100 % loading at 0.8, 0.86 and 0.9 pf, the regulation will be per tabulation given below.

Transformer Power factor	Percentage Regulation			
	100% loading	75% loading	66.67 loading	50% loading
0.8 pf	17	12.53	11.15	8.2
0.86 pf	15.16	9.9	9.71	7.42
0.9 pf	13.56	9.8	8.65	6.3

Conclusion:

The above sizing calculations provide evidence that the Transformer Selected OLTC range for 60MVA Power transformer of -18.59 % to + 10.01 % in steps of 1.43%, range + 7 to -13 taps are adequate as the regulation at extreme negative tap surpasses the calculated minimum regulation for 0.8 pf at 100 % loading. Assuming that the required symmetrical short circuit rating for proposed 13.8 kV Switchgear is 25 kA, with Two Transformers operating in parallel, the estimated total short circuit current at the lowest tap shall be 22.32 kA which is still below the withstand fault current rating of the proposed 13.8 kV Switchgear. Further, the voltage at highest tap is meeting and within the maximum system voltage specified in 01-TMSS-01, IEC-60076-1, IEC 62271-1 and IEC 60038.

3.0 BIBILIOGRAPHY

3.1 01-TMSS-01 Rev. 01, General Requirements for all Equipment/Materials.

3.2 53-TMSS-01 Rev.0, Power Transformers rated above 2MVA up to 100MVA.

3.3 53-TMSS-02 Rev.0, Power Transformers rated above 100 MVA.

3.4 IEC 60038, IEC standard voltages.

3.5 IEC 60076-1, Power Transformers Part 1: General.

3.6 IEC 60076-8, Power Transformers – Application Guide.

3.7 IEC 60214, Tap Changers.

3.8 IEEE C.57.12.90, Standard Test Code for Liquid-Immersed Distribution, Power and Regulating Transformers.